CV – MAX MELCHING

Personal Information

Email	maxm.melching@gmail.com
GitHub	MaxMelching
Website	https://maxmelching.github.io

Education

09/2025 – present — *PhD (Physics)* — California Institute of Technology

10/2022 – 05/2025 — *M.Sc. (Physics)* — Leibniz University Hannover

Final grade: 1,0 (with honors) ✦ Equivalent GPA: 4.0

Minor: Mathematics

10/2018 – 10/2022 — *B.Sc. (Physics)* — Leibniz University Hannover

Final grade: 1,0 (with honors) ✦ Equivalent GPA: 4.0

Minor: Computer Science

Work Experience, Internships

10/2024 – 08/2025 — *Student Assistant* — Max Planck Institute for Gravitational Physics

Various tasks related to the Max Planck Leadnet, such as website development and maintenance, and organizational duties.

Reference: Frank Ohme ✦ frank.ohme@aei.mpg.de

04/2023 – 07/2024 — *Student Research Assistant* — Institute for Quantum Optics, Hannover

Data analysis and software development for the Cold Atom Lab experiment onboard the ISS. Part of that was the development of a Python package that can be used to read, process and evaluate experimental data.

References: Naceur Gaaloul ✦ gaaloul@iqo.uni-hannover.de, Gabriel Müller ✦ g.mueller@iqo.uni-hannover.de

10/2021 – 02/2022 — *Student Assistant* — Leibniz University Hannover

I was one of the tutors for an introductory physics lecture for first-semester undergraduate students. Responsibilities included grading exercise sheets and exams, as well as teaching tutorials to part of the class.

Reference: Tammo Block ✦ block@maphy.uni-hannover.de

05/2026 – 08/2026 — *LSC Fellowship* — LIGO Livingston Observatory

Work as an LSC fellow on site of the LIGO Livingston detector. I participated in several measurements of detector systems, for instance to gather statistics on the performance of the photon calibrator system (performing repeated standard measurements characterizing the system), and contributed to numerous codes related to calibration of the LIGO detectors.

Reference: Brian O'Reilly ✦ brian.oreilly@caltech.edu, Joseph Betzwieser ✦ josephb@caltech.edu

Publications & Theses

2026 *Paper (soon to be published)* arXiv:2604.21859

Title: Mitigating Systematic Errors in Parameter Estimation of Binary Black Hole Mergers in O1-O3 LIGO-Virgo Data

Description: Applies the method developed in the "Accounting for the known unknows" paper to binary black hole mergers that showed systematic differences in their source parameters when analyzed with different waveform models. We specifically look at the events from the first three LIGO-Virgo-KAGRA observing runs (O1-O3).

Author List: Sumit Kumar, Frank Ohme, Max Melching, Harsh Narola, Tom Dooney, Chris Van Den Broeck

2025 *Paper (published in PRD)* arXiv:2502.17400

Title: Accounting for the known unknowns: A parametric framework to incorporate systematic waveform errors in gravitational-wave parameter estimation

Description: Proposes a new way to account for systematic errors in gravitational waveform models and thereby also mitigate the associated biases.

Author List: Sumit Kumar, Frank Ohme, Max Melching

11/2023 – 05/2025 *Master Thesis* Max Planck Institute for Gravitational Physics

Title: Systematic Errors in Gravitational Waveform Models

Description: Theoretical development and extension, as well as implementation, of tools in the Fisher matrix framework that aims at quantifying waveform systematics.

Supervisors: Frank Ohme ✦ frank.ohme@aei.mpg.de, Krishnendu NV ✦ krishnendu.nv@icts.res.in

DOI: 10.15488/19415

03/2022 – 10/2022 *Bachelor Thesis* Max Planck Institute for Gravitational Physics

Title: Systematic Differences in the Source Properties of the Third Gravitational-Wave Catalog

Description: Analysis of the posterior distributions of the third catalog of gravitational wave events, with an emphasis on waveform model systematics.

Supervisor: Frank Ohme ✦ frank.ohme@aei.mpg.de

DOI: 10.15488/19414

Honors & Awards

2023/2024 Deutschlandstipendium ✦ Leibniz University Hannover
(monthly stipend for two semesters)

2022/2023 Deutschlandstipendium ✦ Leibniz University Hannover

(monthly stipend for two semesters)

2020 Niedersachsenstipendium ✦ Leibniz University Hannover
(one-time payment stipend)

Other Information

Computer Skills

Advanced: Python, LaTeX, `git` (includes GitLab, GitHub), Jupyter

Intermediate: Linux, Mathematica, C, Microsoft Windows

Basic: `slurm`, `condor`, MATLAB, Microsoft Office

Presentations

09/2024 Poster "Using Correlations for Good – Systematic Errors Using Alignment" at the LIGO-Virgo-KAGRA Meeting in Barcelona (LIGO DCC: G2401544)

03/2024 Oral Presentation "Systematic Differences in the Source Properties Of Gravitational Wave Signals" at the DPG Spring Meeting of the German physical society, section Gravitation and Relativity

04/2024 Oral Presentation "Systematic Errors in Gravitational Waveform Models" at the DPG Spring Meeting of the German physical society, section Gravitation and Relativity

Personal

Languages: German (Native) ✦ English (Full Working Proficiency)

August 29, 2026